# Omar Abdelhamid

**AI Systems & LLM Engineer** | Agentic Workflows, Advanced RAG & Distributed Serving

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## PROFESSIONAL SUMMARY

Results-driven **AI Systems & LLM Engineer** with a strong engineering and research record across selective telecom R&D labs, prestigious national AI fellowships (DEBI, NTI), and **4x AWS + Azure Certifications**. Proven expertise in architecting production-grade multi-agent topologies (LangGraph), engineering hybrid retrieval backends (Dense + GraphRAG), and orchestrating high-concurrency model serving (vLLM, Triton) on Kubernetes and AWS EKS. Currently pursuing an M.Sc. in Artificial Intelligence while leading applied research on Code SLM reasoning vs. shortcut memorization (*Inv-GRPO*) at Orange Innovation Labs. Dedicated to translating complex foundation model research into deterministic, high-throughput enterprise infrastructure.

## CORE TECHNICAL SKILLS

- **LLMs, Post-Training & Alignment:** SFT, RLHF, RLVR, GRPO, Inv-GRPO, DPO, SimPO, KTO, PPO, LoRA, QLoRA, PEFT, Unsloth, TRL, Model Distillation, Quantization (AWQ, GPTQ, BitsAndBytes NF4), Chain-of-Thought (CoT).
- **Agentic AI Systems & Orchestration:** LangGraph, LangChain, Multi-Agent Topologies, ReAct Loops, Tool/Function Calling, Reflection Nodes, Self-Correction, Human-in-the-Loop (AsyncRedisSaver), Semantic Caching.
- **Knowledge Retrieval & RAG:** Hybrid Retrieval (Dense Vector + BM25), GraphRAG (Neo4j), Vector DBs (Qdrant, FAISS, Pinecone, Milvus), Cross-Encoder Reranking, ColPali Multimodal, Reciprocal Rank Fusion (RRF).
- **Inference Serving & Systems:** vLLM, PagedAttention, Continuous Batching, NVIDIA Triton Inference Server, TensorRT-LLM, ONNX Runtime, gRPC Proxies, FastAPI (Asyncio), KV-Cache Optimization, FlashAttention.
- **Distributed MLOps & Cloud:** Kubeflow (KFP v2), MLflow, Distributed Training (DDP, FSDP, DeepSpeed), Docker, Kubernetes (K8s, EKS), Terraform (IaC), AWS (SageMaker, EC2, S3, RDS, Lambda), Azure AI Foundry, Prometheus, Grafana.
- **Backend Engineering & Fundamentals:** Python (Asyncio), C++ (ICPC Competitive Programming), Celery, Redis, Pydantic, PostgreSQL, Neo4j, Linux/Bash, Git/CI-CD, Clean Architecture.

## WORK & RESEARCH EXPERIENCE

**Orange Innovation Egypt** **Jul 2026 – Present**
*AI R&D Engineer (Applied Research Track)* *Cairo, Egypt*

– Directing applied research on Code SLM reasoning vs. shortcut memorization under Dr. Ghada Soliman; formulated the 6-tier *Reduction Ladder* diagnostic and designed the novel *Inv-GRPO* mitigation algorithm [Codebase].
– Benchmarking open-weight LLMs, RLVR alignment, and multi-agent reasoning workflows for enterprise telecommunications infrastructure, optimizing for deterministic execution, 0ms latency penalty, and ≤8GB edge VRAM constraints.

**Independent AI Systems & Cloud Contractor** **Apr 2025 – Present**
*Lead AI Systems Architect (Remote Commercial Client Engagements)* *Cairo, Egypt*

– **Confidential Client (Enterprise EdTech & LLM Microservices Platform – Ongoing):** Architected a high-concurrency share-nothing microservices cluster using **FastAPI**, unified AI Gateway, and 5 domain-isolated **PostgreSQL** databases; deployed a pedagogical **LangGraph** multi-agent router with **Qdrant** vector search and real-time guardrails (NeMo), fully monitored via **OpenTelemetry, Prometheus, and Grafana**.
– **Confidential Client (Cloud-Native Research Intelligence Platform – 8-Month Delivery):** Architected a collaborative research platform on **Amazon EKS** provisioned via **Terraform (IaC)** and Jenkins CI/CD; built real-time **WebSockets** synchronization and decoupled document indexing via **Celery + Redis**, integrating **pgvector** for dense semantic search and 3D knowledge graphs (**Three.js**).

**National Telecommunication Institute (NTI)** **Jan 2026 – April 2026**
*Agentic AI Systems Engineer (Applied Fellowship)* *6th of October, Egypt*

– Engineered an enterprise-grade LangGraph multi-agent pipeline leveraging ReAct reasoning loops, integrated with ColPali for robust multimodal document understanding and table extraction.
– Architected a resilient 4-layer backend featuring stateful HITL checkpoints via AsyncRedisSaver, automated self-healing SQL query generation, and end-to-end system telemetry (Prometheus/Grafana).

**Orange Digital Center Egypt – Level 3** **Nov 2024 – Jan 2025**
*AI Engineer (Advanced Track)* *Cairo, Egypt*

– Engineered and deployed a real-time YOLOv8 pose estimation pipeline on constrained edge hardware, sustaining 30 FPS inference rate for live interactive feedback.
– Developed deep generative augmentation pipelines (DCGAN, VAE, WGAN) boosting fidelity by 40% FID, and fine-tuned BERT transformer embeddings achieving 92% classification accuracy.

**Digital Egypt Builders Initiative (DEBI)** **Jun 2024 – Oct 2024**
*AWS Machine Learning Fellow (MCIT National Scholarship)* *Cairo, Egypt*

– Provisioned scalable MLOps pipelines on AWS SageMaker, integrating MLflow and Hugging Face to standardize model registries, hyperparameter tracking, and team-wide governance.
– Automated the end-to-end ML lifecycle from distributed multi-GPU training (DDP) to highly available REST API endpoints on AWS, cutting deployment turnaround time significantly.

**IT Gate Academy & Cloud Track** **Dec 2023 – Sep 2024**
*DevOps & Cloud Infrastructure Engineer (Practical Track)* *Cairo, Egypt*

– Orchestrated automated CI/CD pipelines (Jenkins, GitHub Actions) and Terraform (IaC) provisioning for Kubernetes workloads, guaranteeing zero-downtime rolling updates and minimal operational overhead.

## RESEARCH & KEY ENGINEERING PROJECTS

**Reduction Ladder: Diagnosing & Resolving Code SLM Shortcut Mimicry** *Applied Research | Orange Innovation Labs*

- **Research Problem & Diagnostic Suite:** Investigated template memorization versus algorithmic reasoning in Code SLMs; established the 6-tier *Reduction Ladder* diagnostic (664 tasks across HumanEval & EvoEval) evaluating collapse bounds under semantics-preserving AST mutations.
- **Novel Inv-GRPO Formulation:** Derived *Invariance-Regularized Policy Optimization*, pairing cross-perturbation rollouts with semantic consistency rewards ($\mathcal{R}_{\text{cons}}$) to break shortcut mimicry and shift collapse bounds ($\Delta\ell^* \geq 2$) with **0ms inference latency penalty**.
- **Empirical Suite & Profiling:** Benchmarked multi-arm mitigations (Contrastive SFT, AST-RL, Step-RLVR) on Qwen2.5-Coder (1.5B/7B) under edge VRAM budgets ($\leq$8GB); tracked token efficiency, overthinking tax, and error taxonomies.

**NT-sMoCo: Dense Retrieval & Contrastive KBQA** | **NAACL'22 Paper** *Paper Extension | M.Sc. Research*

- **Paper Reproduction & Novel Extension:** Re-implemented and extended the NAACL 2022 paper (*Liu et al.*), formulating a Normalized-Temperature Scaled Momentum Contrastive framework to address entity sparsity in incomplete Knowledge Bases (simulating 30–100% Freebase coverage).
- Engineered a Dual-BERT dense retrieval backbone with hard-negative mining, integrated with a 2-layer Graph Attention Network (GATv2) for multi-hop subgraph entity traversal.
- Benchmarked on WebQSP, successfully replicating published state-of-the-art results (**+7.9% Hits@1** over baseline) with rigorous statistical significance validation ($p < 0.0001$).

**OpenQ: Autonomous Enterprise Agentic Intelligence Platform** *Distributed Agentic Platform*

- Developed an enterprise-ready autonomous platform unifying 8 heterogeneous data modalities behind a natural language interface, eliminating manual BI engineering bottlenecks.
- Architected a hybrid retrieval backend (Neo4j FastRP for structural traversal + Qdrant for dense semantic search), orchestrating asynchronous task queues across 8 fault-isolated Celery workers.
- Implemented a self-correcting LangGraph state machine with 3-retry SQL auto-correction and persistent HITL checkpoints (AsyncRedisSaver), deployed on AWS EKS with Aurora Serverless v2.

**Arabic Mental Health LLM – Distributed Alignment & High-Throughput Serving** *MLOps & LLM Serving*

- Engineered an automated Kubeflow (KFP v2) DAG orchestrating headless execution via Papermill with full MLflow lineage from dataset validation to deployment.
- Fine-tuned Qwen2.5-3B via Unsloth (NF4 4-bit). Benchmarked 6 parallel alignment algorithms (DPO, IPO, KTO, ORPO, SimPO, RLOO), auto-merging the winning SimPO model in BFloat16.
- Configured a high-throughput serving layer using FastAPI and gRPC proxying to NVIDIA Triton + vLLM (PagedAttention, Continuous Batching), maximizing GPU utilization with low-latency API response times.

**Multilingual RAG Architecture with Multi-Agent Reflection** *Enterprise RAG*

- Designed a 5-agent LangGraph topology (Query Rewriter, ReAct Brain, Reflection Node) utilizing LLaMA-3.3-70B to iteratively detect and eliminate LLM hallucinations, ensuring verified factual grounding.
- Optimized retrieval latency using a 5-stage FAISS HNSW pipeline (BM25 hybrid search, cross-encoder reranking, and Redis semantic caching), reducing compute overhead while improving relevance.

**LoRA-Finetuned E-Commerce Semantic Search** *Semantic Search Systems*

- Fine-tuned dual-encoder bi-encoder embeddings via LoRA ($\sim$90% parameter reduction) with FAISS vector indexing, achieving sub-second dense semantic retrieval across 1M+ catalog SKUs.
- Engineered a stateful LangGraph conversational workflow orchestrating FLAN-T5 for context-aware recommendations and personalized multi-turn search sessions.

## CERTIFICATIONS & HONORS

- **AWS Certifications (4x):** Solutions Architect – Associate (Jul 2026), Machine Learning Engineer – Associate (Jan 2026), AI Practitioner (Sep 2025), Cloud Practitioner (Jun 2025).
- **Microsoft Certification:** Azure AI Apps and Agents Developer – Associate (AI-103) (Sep 2026).
- **ICPC Competitive Programming: ICPC ECPC Finalist** – Ranked 4th at Arab Open University, International Collegiate Programming Contest (ECPC Qualifications 2024).
- **Continuous Professional Learning:** Deep Learning, GenAI & NLP (39 courses, DeepLearning.AI); AI Agents & DevOps (13 courses, IBM); MLOps & Kubernetes (KodeKloud).

## EDUCATION

**Benha University – Faculty of Computers & Artificial Intelligence** **Mar 2026 – Present**
*M.Sc. in Artificial Intelligence* *Cairo, Egypt*

**Arab Open University** **2020 – 2024**
*B.Sc. in Computer Science & Artificial Intelligence* *Cairo, Egypt*